



API Consume



Create MVC project

Create a new project

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C# All platforms All project types

Console App

A project for creating a command-line application that can run on .NET on Windows, Linux and macOS

C# Linux macOS Windows Console

Blazor Web App

A project template for creating a Blazor web app that supports both server-side rendering and client interactivity. This template can be used for web apps with rich dynamic user interfaces (UIs).

C# Linux macOS Windows Blazor Cloud Web

.NET Aspire Starter App

A project template for creating a .NET Aspire app with a Blazor web frontend and web API backend service, optionally using Redis for caching.

C# .NET Aspire API Blazor Cloud Common Service Web Web API

ASP.NET Core Web App (Razor Pages)

A project template for creating an ASP.NET Core application with example ASP.NET Core Razor Pages content

C# Linux macOS Windows Cloud Service Web

ASP.NET Core Web API

A project template for creating a RESTful Web API using ASP.NET Core controllers or minimal APIs, with optional support for OpenAPI and authentication.

C# Linux macOS Windows API Cloud Service Web Web API

ASP.NET Core Web API (native AOT)

A project template for creating a RESTful Web API using ASP.NET Core minimal APIs published as native AOT.

C# Linux macOS Windows API Cloud Service Web Web API

Class Library

A project for creating a class library that targets .NET or .NET Standard

C# Android Linux macOS Windows Library

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Create MVC project

Configure your new project

ASP.NET Core Web App (Model-View-Controller)

C#

Linux

macOS

Windows

Cloud

Service

Web

Project name

Address_Consume

Location

E:\2025-2026\Project\StudentsHandsOn

Solution name ⓘ

Address_Consume

☒ Place solution and project in the same directory

Project will be created in "E:\2025-2026\Project\StudentsHandsOn\Address_Consume\"

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Create MVC project

Additional information

ASP.NET Core Web App (Model-View-Controller)

C# Linux macOS Windows Cloud Service Web

Framework ⓘ

.NET 8.0 (Long Term Support)

Authentication type ⓘ

None

☒ Configure for HTTPS ⓘ

☐ Enable container support ⓘ

Container OS ⓘ

Linux

Container build type ⓘ

Dockerfile

☐ Do not use top-level statements ⓘ

☐ Enlist in .NET Aspire orchestration ⓘ

Aspire version ⓘ

9.0

Back

Create

Add UserModel in models folder

```
1  using System.ComponentModel.DataAnnotations;
2
3  namespace Address_Consume.Models
4  {
5      10 references
6      public class UserModel
7      {
8          7 references
9          public int UserId { get; set; }
10
11          [Required(ErrorMessage = "Username is required")]
12          4 references
13          public string Username { get; set; }
14
15          [Required(ErrorMessage = "Email is required")]
16          [EmailAddress(ErrorMessage = "Invalid Email")]
17          [RegularExpression(
18              @"^[^@\s]+@[^@\s]+\.[^@\s]+$",
19              ErrorMessage = "Invalid Email")]
20          4 references
21          public string Email { get; set; }
22
23          [Required(ErrorMessage = "Password is required")]
24          4 references
25          public string Password { get; set; }
26
27          1 reference
28          public DateTime CreationDate { get; set; }
29
30          0 references
31          public DateTime ModificationDate { get; set; }
32      }
33  }
```

Add code in User Controller

► Install package Newtonsoft.Json and add below code in controller

```
Address_Consume
Address_Consume.Controllers.UserController
_client

1  using Address_Consume.Models;
2  using Microsoft.AspNetCore.Mvc;
3  using Newtonsoft.Json;
4  using System.Text;
5
6  namespace Address_Consume.Controllers
7  {
8      1 reference
9      public class UserController : Controller
10     {
11         private readonly HttpClient _client;
12
13         0 references
14         public UserController(IHttpClientFactory httpClientFactory) ...
15
16         0 references
17         public async Task<IActionResult> Index() ...
18
19         0 references
20         public async Task<IActionResult> Delete(int id) ...
21
22         0 references
23         public async Task<IActionResult> AddEdit(int? id) ...
24
25         [HttpPost]
26         0 references
27         public async Task<IActionResult> AddEdit(UserModel user) ...
28
29     }
30
31 }
```

Add code in User Controller

- Install package Newtonsoft.Json and add below code in controller

```
public class UserController : Controller
{
    private readonly HttpClient _client;

    0 references
    public UserController(IHttpClientFactory httpClientFactory)
    {
        _client = httpClientFactory.CreateClient();
        _client.BaseAddress = new Uri("https://localhost:7093/api/");
    }
}
```

Curl

```
curl -X 'GET' \
  'https://localhost:7093/api/User' \
  -H 'accept: text/plain'
```

Request URL

```
https://localhost:7093/api/User
```

Change port no. (API project is running on)

Add GetAll code in U Controller

0 references

```
public async Task<IActionResult> Index()
{
    try
    {
        var response = await _client.GetAsync("User");
        response.EnsureSuccessStatusCode();

        var json = await response.Content.ReadAsStringAsync();
        var list = JsonConvert.DeserializeObject<List<UserModel>>(json);
        return View(list);
    }
    catch (Exception ex)
    {
        TempData["Error"] = "Unable to load users.";
        return View(new List<UserModel>());
    }
}
```


Add Delete code in User Controller

0 references

```
public async Task<IActionResult> Delete(int id)
{
    try
    {
        var response = await _client.DeleteAsync($"User/{id}");
        response.EnsureSuccessStatusCode();
    }
    catch (Exception ex)
    {
        TempData["Error"] = "Unable to delete user.";
    }

    return RedirectToAction("Index");
}
```

Request URL

<https://localhost:7093/api/User/7>

Add Add Edit Form code in User Controller

0 references

```
public async Task<IActionResult> AddEdit(int? id)
{
    try
    {
        UserModel user = new UserModel();

        if (id != null)
        {
            var response = await _client.GetAsync($"User/{id}");
            if (!response.IsSuccessStatusCode)
            {
                TempData["Error"] = "User not found.";
                return RedirectToAction("Index");
            }

            var json = await response.Content.ReadAsStringAsync();
            user = JsonConvert.DeserializeObject<UserModel>(json);
        }

        return View(user);
    }
    catch (Exception ex)
    {
        TempData["Error"] = "Unable to load form.";
        return RedirectToAction("Index");
    }
}
```

Request URL

<https://localhost:7093/api/User/7>

Add Insert and Update code in User Controller

[HttpPost]

0 references

```
public async Task<IActionResult> AddEdit(UserModel user)
{
    if (!ModelState.IsValid)
        return View(user);

    try
    {
        var content = new StringContent(JsonConvert.SerializeObject(user),
            Encoding.UTF8, "application/json");

        if (user.UserId == 0)
        {
            var response = await _client.PostAsync("User", content);
            response.EnsureSuccessStatusCode();
        }
        else
        {
            var response = await _client.PutAsync($"User/{user.UserId}", content);
            response.EnsureSuccessStatusCode();
        }

        return RedirectToAction("Index");
    }
    catch (Exception ex)
    {
        TempData["Error"] = "Unable to save user.";
        return View(user);
    }
}
```

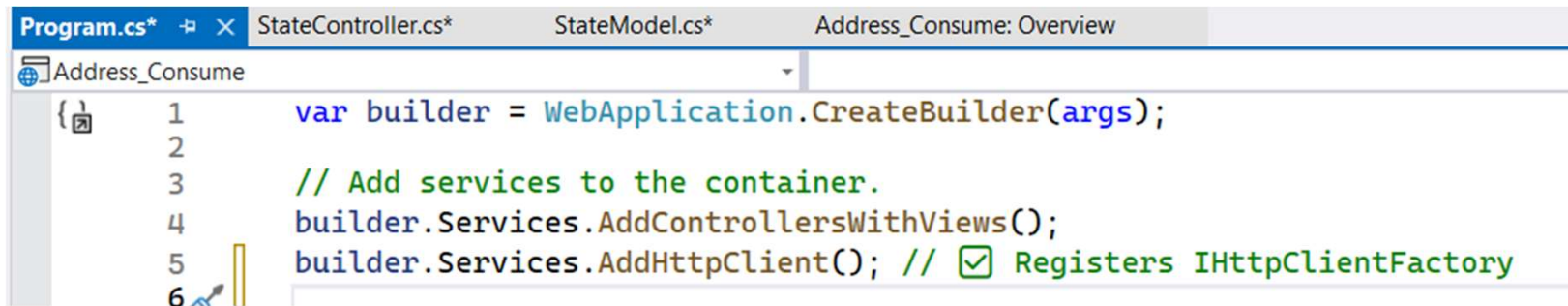
Request URL

<https://localhost:7093/api/User>

Request URL

<https://localhost:7093/api/User/7>

Configure IHttpConnectionFactory in Program.cs



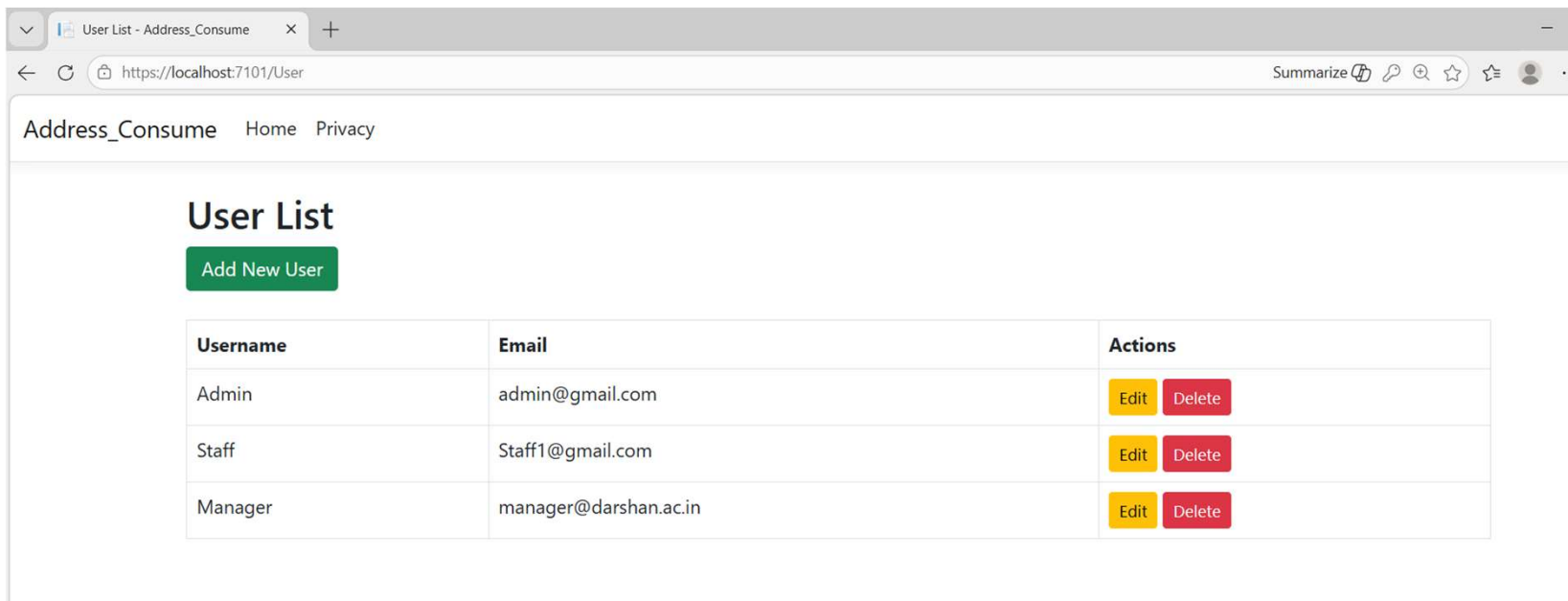
```
Program.cs* StateController.cs* StateModel.cs* Address_Consume: Overview
Address_Consume
1 var builder = WebApplication.CreateBuilder(args);
2
3 // Add services to the container.
4 builder.Services.AddControllersWithViews();
5 builder.Services.AddHttpClient(); // ☒ Registers IHttpConnectionFactory
6
```

Add code in index.cshtml file

```
1  @model List<UserModel>
2  @{
3      ViewData["Title"] = "User List";
4  }
5
6  <div class="container mt-4">
7      <h2>User List</h2>
8      <a asp-action="AddEdit" class="btn btn-success mb-3">Add New User</a>
9      <h3 class="text-danger">@TempData["Error"]</h3>
10
11      <table class="table table-bordered">
12          <thead>
13              <tr>
14                  <th>Username</th>
15                  <th>Email</th>
16                  <th>Actions</th>
17              </tr>
18          </thead>
19          <tbody>
20              @foreach (var user in Model)
21              {
22                  <tr>
23                      <td>@user.UserName</td>
24                      <td>@user.Email</td>
25                      <td>
26                          <a asp-action="AddEdit" asp-route-id="@user.UserId" class="btn btn-warning btn-sm">Edit</a>
27                          <a asp-action="Delete" asp-route-id="@user.UserId" class="btn btn-danger btn-sm"
28                              onclick="return confirm('Are you sure?')>Delete</a>
29                      </td>
30                  </tr>
31              }
32          </tbody>
33      </table>
34  </div>
35
```

Run the project

- ▶ First run API project
- ▶ Then run MVC project
- ▶ Check functionality Select all and delete



The screenshot shows a web browser window with the address bar displaying `https://localhost:7101/User`. The page title is "User List - Address_Consume". The navigation bar includes "Address_Consume", "Home", and "Privacy". The main content area features a "User List" heading and an "Add New User" button. Below this is a table with three rows of user data.

Username	Email	Actions
Admin	admin@gmail.com	Edit Delete
Staff	Staff1@gmail.com	Edit Delete
Manager	manager@darshan.ac.in	Edit Delete

Add AddEdit View page

```
1  @model UserModel
2  @{
3      ViewData["Title"] = Model.UserId == 0 ? "Add User" : "Edit User";
4  }
5
6  <div class="container mt-4">
7      <h2>@ViewData["Title"]</h2>
8      <h3 class="text-danger">@TempData["Error"]</h3>
9      <form asp-action="AddEdit" method="post">
10         @Html.ValidationSummary(true, "", new { @class = "text-danger" })
11         <input type="hidden" asp-for="UserId" />
12         <input type="hidden" asp-for="CreationDate" />
13
14         <div class="mb-3">
15             <label asp-for="UserName" class="form-label"></label>
16             <input asp-for="UserName" class="form-control" />
17             <span asp-validation-for="UserName" class="text-danger"></span>
18         </div>
19
20         <div class="mb-3">
21             <label asp-for="Email" class="form-label"></label>
22             <input asp-for="Email" class="form-control" />
23             <span asp-validation-for="Email" class="text-danger"></span>
24         </div>
25
26         <div class="mb-3">
27             <label asp-for="Password" class="form-label"></label>
28             <input asp-for="Password" class="form-control" type="password" />
29             <span asp-validation-for="Password" class="text-danger"></span>
30         </div>
31
32         <button type="submit" class="btn btn-primary">@((Model.UserId == 0) ? "Add" : "Update")</button>
33         <a asp-action="Index" class="btn btn-secondary">Back</a>
34     </form>
35 </div>
36 @section Scripts {
37     @{
38         await Html.RenderPartialAsync("_ValidationScriptsPartial");
39     }
40 }
41
```

Run both API and MVC project Again

Address_Consume Home Privacy

Add User

UserName

Email

Password

[Add](#) [Back](#)

Address_Consume Home Privacy

User List

[Add New User](#)

Username	Email	Actions
Admin	admin@gmail.com	Edit Delete
Staff	Staff1@gmail.com	Edit Delete
Manager	manager@darshan.ac.in	Edit Delete
NewUser	newuser@gmail.com	Edit Delete

Data added successfully

Run both API and MVC project Again

[Address_Consume](#) [Home](#) [Privacy](#)

Edit User **Update Data**

UserName

NewUser2

Email

newuser2@gmail.com

Password

[Update](#)

[Back](#)

Browser window showing the User List page:

Address_Consume [Home](#) [Privacy](#)

User List

[Add New User](#)

Username	Email	Actions
Admin	admin@gmail.com	Edit Delete
Staff	Staff1@gmail.com	Edit Delete
Manager	manager@darshan.ac.in	Edit Delete
NewUser2	newuser2@gmail.com	Edit Delete

Updated data successfully